

Content Type: Standard Color
Paper Type: White
Page Count: 614
ISBN: 979-8-1827-2111-0
Trim Size: 8.5x11
File Type: PDF
Request ID: CSS5404088
Bleed Artwork Dimensions:
18.489 x 11.250 * 285.8mm x 469.6mm



Algorithms for the problems that matter

Advanced Computational Algorithms provides a rigorous yet practical guide to the design, analysis, implementation, and evaluation of modern algorithms.

The Second Edition develops mathematical foundations while connecting theory to executable implementations and real-world computational problems. Topics include asymptotic analysis, divide-and-conquer methods, advanced data structures, greedy and dynamic programming techniques, randomized and approximation algorithms, computational complexity and NP-completeness, graph algorithms, string processing, numerical methods, and modern algorithm engineering.

Worked examples, pseudocode, executable code, instructional diagrams, proofs, exercises, benchmarking activities, and applied projects help readers move from understanding algorithms to evaluating how they behave in practice.

Designed for advanced undergraduate and graduate students, instructors, software engineers, data scientists, researchers, and technical professionals, the book emphasizes correctness, complexity, performance evaluation, reproducibility, and computational problem solving.

ABOUT THE AUTHOR

Moody Amakobe is a multidisciplinary researcher, data scientist, and blockchain architect whose work spans artificial intelligence, computational systems, data engineering, and distributed architectures. As founder of the Global Data Science Institute, he leads work in data science, deep learning, algorithm design, and technology for impact. He has taught at multiple universities, authored technology textbooks, advised research teams, supervised graduate projects, and developed industry solutions across healthcare, finance, agriculture, and public systems.

OPEN EDUCATIONAL RESOURCE · CC BY 4.0
Designed to support accessible teaching, independent study, and lifelong learning.

Published by Global Data Science Institute



ADVANCED COMPUTATIONAL ALGORITHMS • MOODY AMAKOBÉ

1.239
31.47mm

Template Instructions:

- **Pink (Safe) Area:** All text, important images, logos and barcode should remain inside the pink areas.
- **Blue (Bleed) Area:** All backgrounds should extend throughout the blue and pink areas. If a white background is desired, create a white box that covers the entire bleed area.
- **White (Template) Area:** No artwork should extend into the white areas that contain the black text and fold marks and template info. Please leave all elements outside of the pink and blue areas unaltered.
- **Barcode Placement:** The provided barcode may be moved to any location within the pink area of the back cover.
- **Document Size:** The overall document size for this template is shown in the lower left corner. The final cover PDF submitted must match this size.

Pink and blue boxes must be completely covered when cover is submitted.

GLOBAL DATA SCIENCE INSTITUTE

8.50 x 11.0
279 mm x 216 mm

8.50 x 11.0
279 mm x 216 mm